

'AI-Teacher' Assistant System: A Smart Attendance and Participation tracking system for students

Mouza Alzaabi¹, Mahra Almeheiri¹, Shuaa Alqubaisi¹, Ahmed Shuhaiber.

¹College of Technological Innovation (CTI), Zayed University, Abu Dhabi, UAE

Emails: 201913282@zu.ac.ae, 201812348@zu.ac.ae, 201908355@zu.ac.ae, Ahmed.Shuhaiber@zu.ac.ae

Abstract— Students and teachers are connected in an orderly way to attain learning outcomes, which is the foundation of attendance systems. Academic institutions, however, encounter poor strategies for these parties' participation. In order to address the low attendance and participation among students and faculty in a university community, a university-wide 'AI-Teacher' system will be built in one of the educational institutions in the Gulf region. The system was prepared for by planning, gathering, modelling, and analyzing the requirements, designing the interfaces and databases, and then coding and testing. The AI-Teacher assistant system is approachable and simple to use. A warning is sent to parents after a student misses a predetermined number of classes, along with an automatic or manual attendance record, alerts, and notifications. Also, the application system, 'AI-Teacher' which works with iOS and Android devices, can improve connections between "students and instructor". The study concluded with some recommendations for further research and development.

Keywords— *AI system, Facial recognition, Performance recognition, Attendance, students, teachers*

I. INTRODUCTION

It is evident based on studies that students' participation and attendance in class increases their motivation, helps their learning, enhances their communication, and encourages higher order thinking skills [1-4]. In order to determine the underlying causes of the current level of university students' participation and attendance in class, this study will look into those elements. In the world we live in today technology has a significant impact on all areas of our daily lives, therefore it has revolutionized the way we live. Technology can also alter the conventional methods of teaching and learning in the academic setting. Therefore, in this paper, we aim to: 1) help instructors in simplifying the process of recording student's attendance and participation records 2) improve the experience of being a student and a professor on campus; 3) make it simple for instructors and guardians of students in obtaining more meaningful information about students' attendance and participation records.

All businesses, government agencies, SMEs, and educational institutions now have their own systems to manage everyday operations due to the quick development of technology [2, 5]. For instance, it has proven difficult for university staff members, including professors, parents, and advisors, to have a simple and straightforward approach to record attendance and monitor student progress. The University

currently uses a limited number of ineffective ways to track attendance. For instance, instructors would manually record attendance on their device calling each student which takes approximately 15% of class time [6]. Another example, students and teachers may encounter delays when utilizing ID scanners to register attendance prior to entering the classroom because of slow system response times, long waiting in lines, and frequent system failure [6,7]. Several universities now also offer these students finger scanning and other solutions. Yet, none are effective and because it is challenging to maintain track of all students in all sections, there isn't an efficient attendance method via which university instructors may receive updates regarding their students' performance either. Moreover, SAP/FAP portals are the only way for students and faculty to receive attendance updates.

To solve these issues, we are introducing a clever solution that enables instructors to maintain strong attendance records. The system will automatically record attendance with the help of Artificial intelligence technology saving allowing instructors and students to save valuable time. Moreover, "AI-Teacher Assistant" system, will send notifications and warnings to guardians of students who skip more than the predefined days of classes or if there are any changes regarding performance. Additionally, the smart system will repeat the process of taking attendance every 20 minutes to guarantee the student is in the classroom for the entirety of the lecture. Furthermore, the solution will allow educators and pupils to benefit by having classes full of engagement by setting up an inviting atmosphere for discussion of topics, assignments, and academic prospects.

In order to achieve this, this article is organized as follows: In Section 2, significant publications from earlier investigations are briefly introduced in terms of scope and methodology. The objectives and study methodology are succinctly outlined in Section 3. The relevance of the study is then presented in Section 4, in both the discussion and the findings. A conclusion is the focus of Section 5. The concluding Section 6 contains the restrictions and future works.

II. LITERATURE REVIEW

Although there have been decades of AI study, the area has just recently experienced substantial advancements [8-11, 15]. The creation of machine learning techniques, which enable computers to learn from enormous datasets, is one of the primary causes of this [9, 14]. AI systems that are capable of a

wide range of activities, including image recognition, natural language processing, and gaming, have been created using these techniques [16].

Convolutional neural networks (CNNs) and recurrent neural networks (RNNs), two models within the deep learning subfield of machine learning, have recently achieved state-of-the-art performance on a variety of tasks [10-13]. These models have been put to use in many different applications, including speech recognition, self-driving automobiles, and image categorization.

A. IBM Watson

Leading AI market competitor IBM Watson offers cutting-edge AI technologies and services to companies and organisations around the world. Deep learning, machine learning, and natural language processing are just a few AI technologies provided by IBM Watson. AI technology of IBM Watson linked with our product will increase the benefits of using it and fusing additional existing technologies.

B. Google Cloud AI Platform

Another significant participant in the AI sector is Google, which provides a potent AI platform that enables companies to create and implement their own AI applications. Pre-trained models, API's, and tools for deep learning and machine learning are available through the Google Cloud AI platform.

C. Microsoft Azure

A variety of AI services, such as machine learning, cognitive services, and bots, are available through Microsoft Azure. Business may create intelligent applications and automate business operations with the use of the tools and services offered by Azure AI.

D. Amazon Web Services

Leading cloud computing and AI services are offered by Amazon Web services. AWS provides a number of AI services, such as speech recognition, picture analysis, and natural language processing.

E. Intel

Leading hardware for AI applications is offered by Intel. The company provides a selection of AI platforms and technologies that let companies develop and deploy AI applications more quickly.

III. METHODOLOGY

A. Planning Phase:

A primary goal is to identify the technologies and tools required for developing the Virtual Assistant Application System. Below, we list the hardware and software components that are most closely related to the virtual assistant app: ardware (Artificial intelligence camera, computer device, and a mobile device), and software (CSS, Adalo, Python, and MySQL). Starting with designing the pages (using the Adalo tool) and entity relationships (using MySQL), we will build the app from the ground up before coding the entire system. The database is then connected to Adalo. The hardware camera is then programmed and connected to the app.

1) Financial Feasibility

Tangible benefits of the App include revenue from sales and cost savings, whereas intangible costs include better user experience, higher customer satisfaction, brand equity, and better compliance.

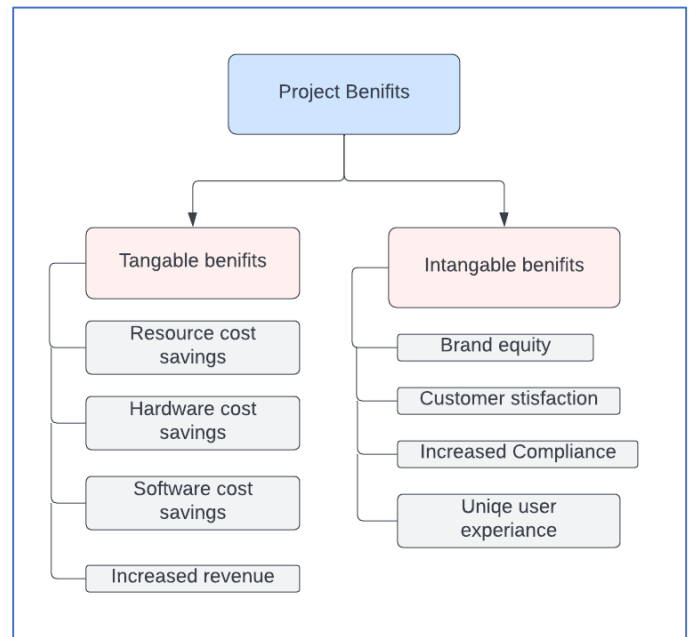


Figure1: Project benefits both (tangible and intangible)

B. Analysis Phase:

1) Requirements gathering

a) Business Requirements

The requirements for this application are as follows; each criterion describes the necessity for this product and how it will benefit the users. The following are the system's business requirements: The application should do the following:

- help instructors manage their time in class well
- help students with attendance and performance
- inform instructors when students left the class for a long period of time
- notify students with their performance in class and attendance
- notify instructors and students when the camera is recording and when it stops recording
- enables parents to get information about their children in class

b) User Requirements

The proposed system will include four key stakeholders: Students, Instructors and Parents. The students will be able to:

- review attendance
- performance record
- receive warning notification on attendance and performance
- upload evidence
- comment on attendance and performance

On the other hand, the *instructors* will be able to:

- login
- input their ID and classes
- start the class
- review the video after class
- writes feedbacks about each student
- reject or accept student attendance record

The *Parents* will be able to:

- Receives notifications when their child's performance is low and absent is high
- be able to communicate with the instructor

c) System Requirements

The system has to have the ability to record classes and alert instructors when students aren't participating or paying attention. A student's parents will be notified about their absence if they have been absent from class three times. There will be a time limit on how many times instructors can watch the recordings.

The system must have the following functions: process functions, input functions, and output functions. The input function will contain the permissions that the application offers, the output will be the application's public display, and the process will be the action that the program takes.

The system will allow input:

- instructors to enter login data
- instructors reviewing each student
- students will review their absent and performance level
- parents receiving notifications
- adding instructor's data and courses by staff

The system will display the follow (output):

- camera recording
- course number and section
- instructor's information
- students' information

The system will perform the following processes:

- saves students information
- sends notifications
- sends alerts to students when their performance level is low and attendance is high
- alerts when it starts recording and ends recording

In terms of performance, the application should:

- operates constantly
- able to accommodate multiple instructors simultaneously
- student's data are secured
- recordings are secured
- only instructors can access the recordings
- students can only upload absence documentation
- staff may add or remove courses

2) Process Modeling

The process modelling shoes the data flow diagram, as in Figure 2.

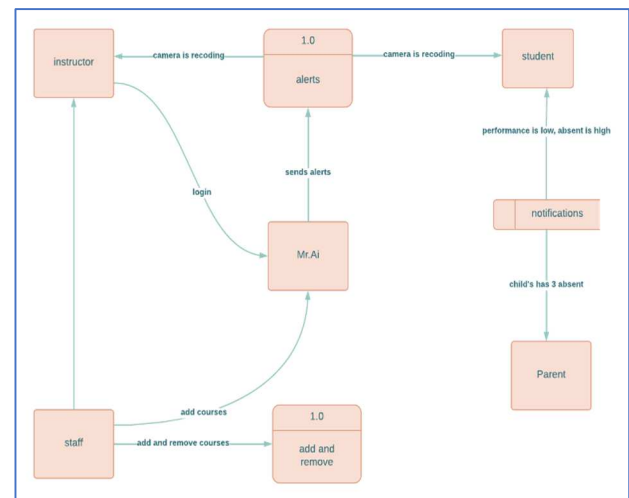


Figure2: Data Flow Diagram

3) Data Modeling

Regarding the data modeling, the entities, their attributes and the associated relationships are demonstrated in order to model and design the database of the application. Therefore, the next figure demonstrates the Entity Relationship Diagram (ERD), which shows five main entities and their attributes.

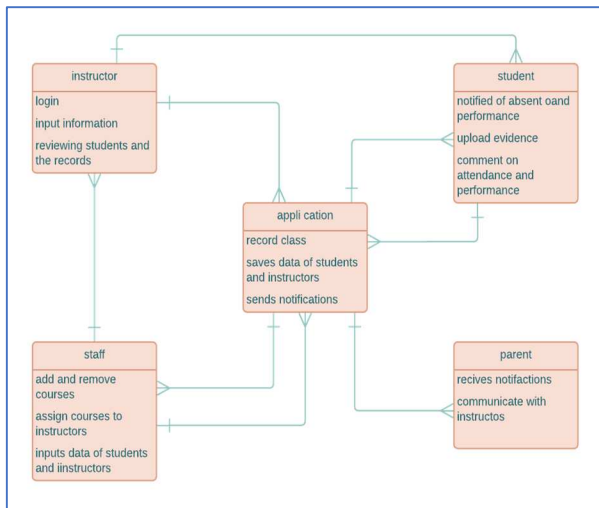


Figure3: Entity Relationship Diagram

In addition, Figure 4 illustrates how the researchers considered modeling the choices and circumstances connected with the system.

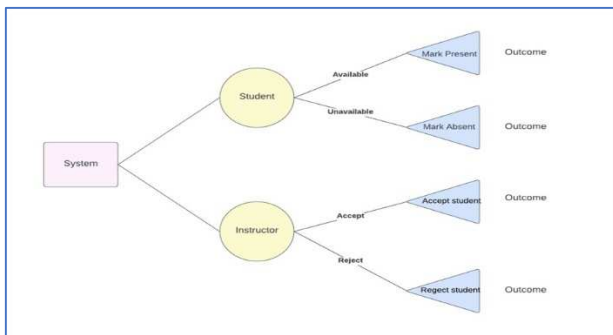


Figure4: Decision Tree (live Chat)

2) User interfaces (GUI)

Below are some figures that demonstrate how the system functions.



Figure5: Facial Recognition

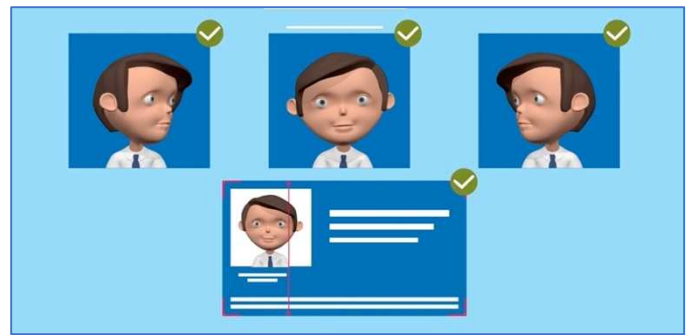


Figure6: Identity authentication

Figure 5 and 6 above demonstrates how the facial recognition tool compares the student's face to their ID photo.



Figure7: Performance recognition

In order to detect students in the detection zone, the artificial intelligence camera will be mounted on top of the whiteboard as shown in the figure 7.

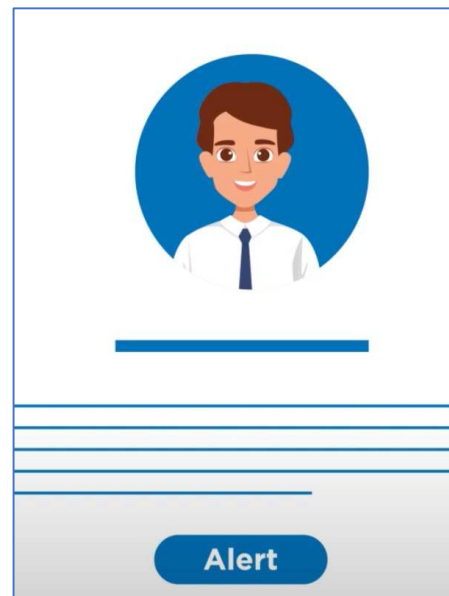


Figure8: Summary report

Figure 8 above shows an alert message which all parties will receive once a student has altered one of the triggers such as being absent.

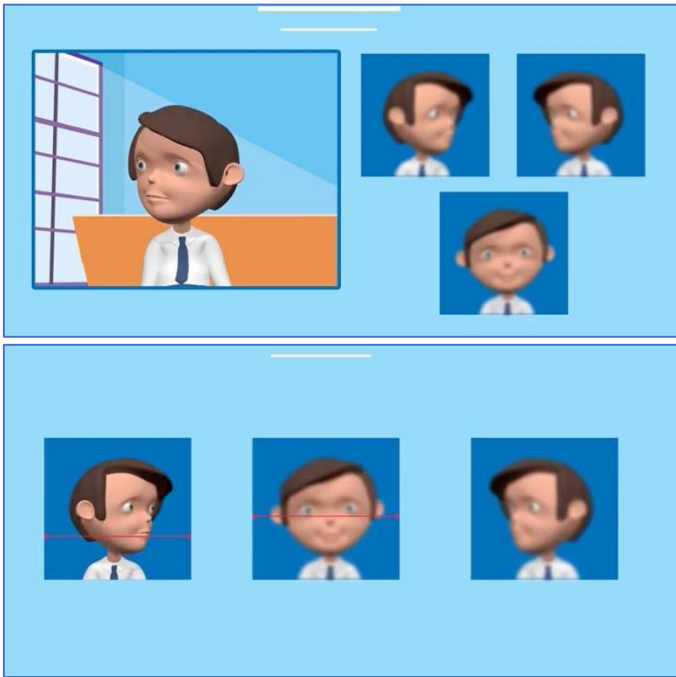


Figure9: Movement and gestures recognition

Figure 9 above shows process of using a cleansing tool to make blurry student photos into High definition.



Figure10: Attendance action

Figure 10 above demonstrates how the instructor can manually view, approve, or disapprove attendance if needed.

IV. DISCUSSION

The attendance and participation issue, which has a significant impact on the instructors' teaching journey and the academic journey of the students, can be resolved with the help of the innovative "AI-Teacher Assistant" system. It also benefits the institution since it gives users—students, faculty members, and parents—access to the crucial data they need to

be informed about events on campus related to student performance. As a result, this paper adds to the body of literature review by offering yet another superb remedy for resolving student issues and improving their on-campus experiences.

The “AI-Teacher Assistant” app claims to bring many benefits to users on campus and guardians. In order to improve the experience and help both sides interact more effectively in class while preventing distractions, the program, for instance, offers a live preview of student conduct for both learners and educators. For instance, if a student's eyes are downward for an extended period of time, the system will notice and the student may be on their phone. In addition, This application gives parents peace of mind and assurance because it contains a function that allows everyone to receive a warning notification if a student misses a predetermined number of lessons, giving instructors and parents the opportunity to discuss the causes and direct the student if needed. Moreover, the system repeats recording attendance every 20 minutes or sets the predefined timings to guarantee the student is in the classroom for the entirety of the lecture. Not only that but the app includes a feature for collecting opinions that could potentially improve the quality and functionalities of teaching the systems based on users comments. The characteristics and scope demonstrated that our system is indeed unique and distinct of its kind.

The opportunity to save valuable class time and make it more beneficial will be one of the many advantages this application will offer its users.From the perspective of the student, the application will enhance their academic experience because they will be able to complete their assignments, pay attention in class, and arrive on time. From the faculty viewpoint, the solution offers a fascinating feature that records attendance and performance for them, allowing them to finish their tasks quicker, wasting no time and eliminating human error. From a parents perspective, It also serves as a means of connection for parents to check on their children at university ensuring everything is going smoothly.

V.CONCLUSION

In this document, we describe the AI-Teacher Assistant system's study, planning, and operation. The AI-Teacher Assistant system helps students, instructors, and parents to easily and simply receive data about students' behaviour in the University and their attendance records, either automatically through a smart system like artificial intelligence or manually by the instructor manually entering the record into the system. To help students have a helpful and joyful environment in the university there are few features we plan to add to our application in the future. For instance, Adding a reward system which will encourage students to participate more and be on time. Moreover, the artificial intelligence system can improve by learning from students' behaviour in future experiences. In addition, based on our customer comments we may Improve the quality and features of our solution.

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